

FinTrust Digital Bank
Week 3 Part 4- Validated Business Findings

Cross-validation of major findings using SQL, Python and Power BI

What this section is

Part 4 validates the major business findings developed during Week 3. The purpose is to check whether conclusions remain consistent when examined through different analytical methods. Each finding is presented with its evidence, the validation method used, the resulting business meaning, and a final status indicating whether the finding is supported or requires revision.

Validation approach

The validation process compared outputs from the advanced SQL analysis, the Week 3 Python analysis and the updated Power BI dashboard. Matching results across tools increase confidence in the finding, while any differences are used to refine or qualify the conclusion rather than overstate it.

Finding 1 - February activity declined, followed by a strong March recovery

Finding	Transaction activity declined in February 2026 and recovered strongly in March 2026.
Evidence	January recorded 4,133 transactions worth approximately NGN 188.49 million. February fell to 3,734 transactions and approximately NGN 175.79 million. March recovered to 4,133 transactions and approximately NGN 196.20 million.
Validation	The SQL monthly trend analysis and Python monthly analysis produced the same monthly transaction counts and values. The Power BI monthly trend also shows the February decline and March recovery.
Business Meaning	FinTrust experienced a temporary slowdown in February. March returned to January transaction volume while producing a higher total transaction value, indicating stronger value performance during the recovery.
Status	SUPPORTED

Finding 2 – Higher monthly income bands receive substantially more risk review

Finding	Higher monthly income bands(250k-499k) are much more likely to be selected for risk review.
Evidence	Risk-review rates increased from 17.66% for transactions below NGN 10,000 to 22.13% for NGN 100,000-199,999 and 40.54% for NGN 200,000 and above. The top 10% highest-value transactions also had a 36.00% review rate compared with 17.78% for the other 90%.
Validation	SQL and Python produced identical amount-band counts and risk-review rates. The updated Power BI Risk Review page reproduces the same pattern and highlights the sharp increase for transactions of NGN 200,000 and above.
Business Meaning	Transaction value is strongly associated with risk-review selection, particularly at higher amounts. The synthetic Risk_Review_Flag represents additional review only and must not be interpreted as confirmed fraud.
Status	SUPPORTED

Finding 3 - Failure rates vary by both banking channel and transaction type

Finding	Transaction failure rates are not evenly distributed; they differ depending on the combination of channel and transaction type.
Evidence	POS Bill Payment recorded the highest failure rate at 7.22%, followed closely by USSD Transfer at 7.20%. Web Card Purchase was lower at 3.56%. Mobile App Transfer had a 6.09% failure rate but 93 failed transactions because of its much higher transaction volume.
Validation	The SQL Channel x Transaction Type analysis matched the Python heatmap. The Power BI failure-rate matrix reproduced the same channel and transaction-type pattern.
Business Meaning	Operational monitoring should consider both failure rate and absolute number of failures. A high percentage identifies failure propensity, while a high failure count may indicate a larger operational impact because of transaction volume.

Status	SUPPORTED
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Finding 5 - Segment totals can hide differences in per-customer performance

Finding	Overall segment size can mask which customer segments perform better on a per-customer basis.
Evidence	Everyday generated the greatest overall activity because it has the largest customer base. After normalization, SME generated the highest transaction value per customer at NGN 387,923.02, while Student recorded the highest transaction frequency at 8.23 transactions per customer.
Validation	The normalized SQL segment analysis was reproduced in Power BI using Transaction Value per Customer and Transactions per Customer measures.
Business Meaning	FinTrust should evaluate segments using normalized measures alongside totals. A large segment may lead in total activity without being the strongest segment on a per-customer basis.
Status	SUPPORTED

Validation Summary

Cross-validation across SQL, Python and Power BI showed strong consistency in the major FinTrust findings. Monthly transaction trends, transaction failure patterns, risk-review behaviour and normalized segment performance were reproduced across the relevant analytical tools.

Overall conclusion: Four of the five major findings were fully supported, while one was revised based on the validation evidence. The validated results provide a reliable basis for the management recommendations developed in Part F.